



**REAL-TIME AI NAVIGATION AND
OBSTACLE DETECTION FOR BLIND
USERS**

A PROJECT REPORT

Submitted by

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*in partial fulfilment for the award of the degree
of*

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

**KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)**

ANNA UNIVERSITY :: CHENNAI 600 025

APRIL 2026



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1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identity, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-Long Learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY
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BONAFIDE CERTIFICATE

Certified that this project report **“REAL-TIME AI NAVIGATION AND OBSTACLE DETECTION FOR BLIND USERS”** is the bonafide work of **NIVETHA V (621322104065), SIVA BHARATHI A (621322104100), TAMIL K (621322104113)** who carried out the project work under my supervision.

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Submitted for the Main project viva-voce examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We wish to express our sincere thanks to our beloved, respectful and honorable Chairman **Dr.PSK.R.PERIASWAMY** for providing immense facilities in our institution.

We proudly render our thanks to our Respected Principal **Dr.R.ASOKAN, M.S.,M.Tech.,Ph.D.**, for the facilities and the encouragement given by him to the progress and completion of our Project

We would like to express our gratitude to our Dean(R&D), **Dr.J.YOGAPRIYA,M.E.,Ph.D.**, and Dean (Academics & IQAC) **Dr.J.PREETHA,M.E.,Ph.D.**, who are the continual source of inspiration to think innovatively to complete the Project

We proudly render our immense gratitude to our Head of the Department of Computer Science and Engineering **Dr.C.SARAVANABHAVAN,M.Tech, Ph.D.**, for his effective leadership, encouragement and guidance in the Project.

We would like to express our heartfelt thanks to our respected supervisor, **Mrs.S.SUGANYA,M.E.,(Ph.D).**, for her invaluable guidance and support throughout the completion of our project.

We highly indebted to provide our heartfelt thanks to our respectful Project Coordinator **Ms.V.DHANALAKSHMI,M.E.**, for her valuable ideas, constant encouragement and supportive guidance throughout the Project.

We wish to extend our sincere thanks to all teaching and nonteaching staffs of Computer Science and Engineering department for their valuable suggestions, cooperation and encouragement on successful completion of this project.

We wish to acknowledge the help received from various departments and various individuals during the preparation and editing stages of the Manuscript.

ABSTARCT

Visually impaired individuals often struggle to navigate safely in outdoor environments because traditional tools and basic navigation apps do not provide enough support for understanding surroundings, identifying obstacles, or handling unexpected situations. Existing solutions are either too simple, require extra devices, or fail to give clear, real-time guidance during walking or while using public transport, which reduces independence and increases the risk of accidents. To address these challenges, the proposed system introduces an easy-to-use web-based assistant that uses simple forms of machine learning and deep learning to guide users through voice instructions, detect obstacles, understand how the user is moving, announce nearby locations and bus stops, support multiple languages, and provide a quick emergency alert to trusted contacts. By working directly through any device's browser, the system removes installation barriers, improves safety and confidence for visually impaired users, and offers a more accessible and reliable travel experience in everyday environments.

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LIST OF ABBREVIATIONS

AI	A rtificial I ntelligence
API	A pplication P rogramming I nterface
BLE	B luetooth L ow E nergy
CNN	C onvolutional N eural N etwork
DL	D ee P L earning
GPS	G lobal P ositioning S ystem
LIDAR	L ight D etection A nd R anging
ML	M achine L earning
OSM	O pen S treet M ap
RF	R andom F orest
SOS	S ave O ur S ouls
SVM	S upport V ector M achine
YOLO	Y ou O nly L ook O nce

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Navigation and mobility solutions for people with visual impairments continue to present a major challenge, especially outdoors in mixed environments such as sidewalks and public transit. There are a lot of obstacles we have to face when moving from point A to B, because we have to be aware of many environmental factors, since finding our path without interruptions is an important issue. Unfortunately, current assistive technology continues to fall short of this demand, merely providing excursions with standard GPS directions and with the ability to detect certain obstacles.

Similarly, like the agricultural system, which also depends on the monitoring and decision-making process, the navigation system also depends on the continuous monitoring of the environment, analysis of the patterns, and guidance systems. There are stages in the travel process. These include the beginning of the trip, the trip to a certain point, crossing the road, taking the bus, and stopping the transport. Each stage has its own kind of supporting systems, which have to be integrated. For the walking stage, the need arises to identify the obstacles and the surfaces, while for the bus, the need arises to identify the stops and alerts, and whether the passenger is supposed to get down at the stop. There is a need to be vigilant and identify the minor risks, such as steps, poles, vehicles, and the route, which might need a revision, and it could also result in a major risk.

It is noted that the integration of Machine Learning (ML), and Deep Learning (DL) technologies, in addition to real-time GPS tracking technology, along with voice interaction technology, has led to the development of the Real- Time AI Navigation and Obstacle Detection System for Blind Users, which is a holistic navigation aid tool that analyses the pattern of the user's navigation to provide navigation guidance.

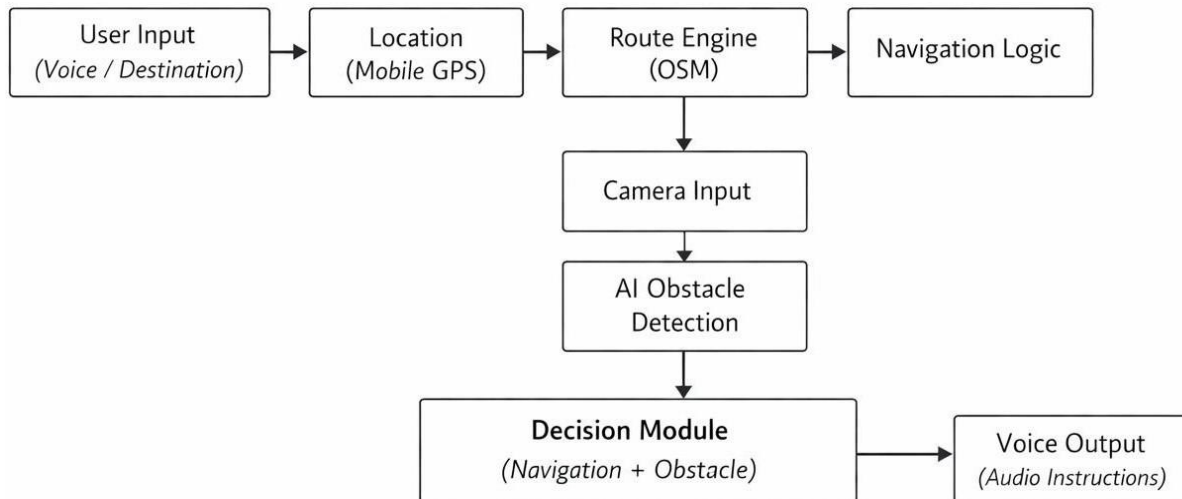


Figure 1.1 Structure of Real-Time AI Navigation and Obstacle Detection System

The system can make use of the smartphone's camera during pedestrian navigation. This system will implement deep learning-based object detection technology to detect obstacles in the pedestrian's way while pedestrian navigation is in progress. Obstacles the system will detect are cars, poles and stairs. The system will notify pedestrians of the obstacles. This system will suggest an alternative path for the pedestrian. The system will alert pedestrians about uneven ground and crowded areas. The system will alert about current bus stops while a pedestrian is on a detected bus. The system will alert about upcoming bus stops and the final bus stop.

Smart Voice Interface-It is an intelligent voice interface through which we can navigate the whole application and dictate the location. Also, it features an adaptable voice interface, due to which it could be used in the open air efficiently. **SOS Function**-An SOS function has been incorporated whereby an individual needs to hold down the button for a while so that a distress message can be sent to a reliable person, so that an awareness about the incident.

Most available assistive solutions have used machine learning for movement detection, deep learning for object recognition and GPS for navigation; however, they focus on just one aspect of mobility. This solution proposes a combination of the following functions: navigation, obstacle detection, public transportation assistance, movement analysis and safety alert to be integrated into a coherent system. The system works with real-time information of the environment coupled with processing power and voice commands to promote safety, independence and confidence of blind users in travelling.

1.2 ARTIFICIAL INTELLIGENCE IN ASSISTIVE TECHNOLOGY

AI has made assistance devices much more efficient; it is now possible for the device to interact with its environment smartly, recognise, and respond to different situations. Assistive devices cannot work and compare the behaviour of traditional assistive devices. However, AI-based devices have been found to recognise, adapt to new conditions, and provide contextualised actions. ML, DL, Computer vision and speech processing techniques can achieve this.

For people who are blind or visually impaired, AI-based systems acting as a guide allow them to gain contextual awareness of their environment and communicate information about the environment to the users using audio. For example, obstacle identification in real time, context-aware turn-by-turn navigation, classification of human movement and classification of environmental sounds increase a user's awareness.

A navigation and obstacle detection integrated system takes motion detection algorithms to detect if the user is walking, stationary or using public transport. Deep learning based YOLOv8 models take input from real-time camera feed and detect obstacles (e.g., car, pole, step, ground not flat) and display instructions with the help of mapping applications like OpenStreetMaps.

With the usage of voice recognition and text-to-speech, totally hands-free operation could be made possible, from speaking the destination address, navigating via turn-by-turn voice, to triggering the emergency calls. Noise filtering techniques improve the quality of speech recognition in a busy outdoor environment, and a rule-based emergency decision system makes sure that the contacts can receive notifications immediately in emergency situations. With the application of perception, prediction and interactive adaptability, the assistive navigation evolves from an assistive tool to an intelligent assistant, which increases the independence, confidence and safe navigation capabilities for the visually impaired.

1.2.1 Applications of AI in Navigation Systems

AI technology in navigation can also take this one step further. Through intelligent decision-making, it will be possible for real-time analysis, learning and adaptation to take place. It is an AI navigation system that actively observes the user and surroundings and takes actions to improve the safety and efficiency of the trip.

- **Intelligent Route Optimisation**

AI studies current real-time GPS feed and available digital maps (eg, OpenStreetMaps), and calculates the best route (shortest, most efficient or safest route). If there is a change in destination, detours or a blockage in the path, the routes are auto-updated.

- **Real-Time Obstacle Detection**

Deep Learning networks like YOLOv8 will perceive the obstacles, such as vehicles, poles, steps, slopes, and pedestrians, from real-time video. These identified obstacles are converted to instantaneous audio warnings, thus avoiding collisions.

- **Movement Detection and Classification**

The ML algorithms would then take the data coming from the motion sensors, and figure out whether I am walking, standing still or driving a car. Then the navigation guidance can be altered.

- **Voice-Based Navigation Control**

AI-fueled speech recognition permits a user to provide commands and receive spoken, turn-by-turn directions entirely by voice.

- **Public Transport Assistance**

AI detects public bus trips based on motion and GPS information, it announces the arrival of stops close by and also reminds the user as they approach the stop of their destination.

- **Contextual Awareness**

Based on contextual inputs from the user's environment, AI can give contextually relevant alerts via spoken word to inform the user of an awareness of these conditions.

- **SOS alert integration**

Upon detecting an SOS alert, the AI provides an alert via spoken word to a trusted group of individuals.

1.3 SDG INTEGRATION IN PROJECT DEVELOPMENT

SDG 7: Affordable and Clean Energy

Efficient energy utilization is supported through analysis of electricity consumption patterns and operational energy usage. The system enables identification of high energy-consuming activities and promotes optimization strategies for reduced energy waste. By providing data-driven insights into energy consumption, it encourages the adoption of cleaner and more sustainable energy practices, contributing to improved energy efficiency and responsible energy management.

SDG 9: Industry, Innovation and Infrastructure

Advanced machine learning techniques and data-driven methodologies are utilized to enhance industrial emission monitoring and analysis. The integration of ensemble regression models demonstrates the application of innovative technologies in environmental management. The system supports the development of intelligent infrastructure capable of handling complex operational data, thereby promoting sustainable industrial innovation and improved analytical capabilities. The system also supports scalability and adaptability for integration with future industrial technologies and smart monitoring systems.

SDG 12: Responsible Consumption and Production

Resource utilization and emission-generating activities are systematically analyzed to promote responsible consumption practices. The system facilitates identification of inefficient processes and high emission sources, enabling optimization of operational activities. By encouraging data-driven decision-making, it supports sustainable production patterns, reduces resource wastage, and enhances overall operational efficiency in alignment with responsible consumption principles. It further promotes efficient resource management by enabling continuous monitoring and optimization of energy and material usage.

SDG 13: Climate Action

Carbon emission prediction and analysis contribute to effective climate change mitigation efforts. The system enables identification of major emission sources and supports monitoring of emission trends over time. These insights assist in developing targeted reduction strategies and informed policy decisions. The analytical approach strengthens environmental accountability and supports proactive measures for reducing the overall carbon footprint. The system also aids in raising awareness and supporting sustainable practices through data-driven environmental insights.

1.4 MOTIVATION OF THE PROJECT

Navigating independently can be a real problem for blind people, as they do not get enough knowledge about their environment and support to overcome it. Basic GPS applications have no facilities for obstacle recognition, motion detection or help in an emergency, as a result which increases reliance on others.

Advancements in AI technologies, combined with Smartphone availability, have enabled an integrated navigation system to be developed. The navigation system will offer: real-time directions, warnings for potential obstructions and voice responses.

1.5 OBJECTIVES

The final system that would be implemented for the "Real-Time AI Navigation and Obstacle Detection for visually disabled people" is an intelligent voice-operated guide for the blind to walk independently and safely.

The key objectives are:

- The aim is to develop a real-time navigation system providing accurate turn- by-turn voice guidance through the integration of GPS and digital maps.
- The implementation of deep learning-based obstacle detection using various models like YOLOv8 for the detection of cars, poles, steps and bumpy terrain.
- To implement a motion detection system to distinguish between activities (walking, waiting and riding a bus) using machine learning.
- Public transport assistance by announcing nearby bus stops and arrival at the user's desired stop.
- Full voice-based interaction through the combination of speech recognition and text-to-speech modules to allow hands-free usage.

CHAPTER 2

LITERATURE REVIEW

[1] TITLE: EFFICIENT REAL-TIME PATHFINDING FOR VISUALLY IMPAIRED INDIVIDUALS (2025)

AUTHOR: Tadeh Ghahremani, Hossein Mahvash Mohammad

To fully benefit from the camera and GPS for blind navigation, and eliminate the shortcomings of the camera and GPS, the main functionalities are: Real-time path planning and avoidance are crucial to ensure secure and flexible individual mobility for the visually impaired. We would like to propose an integrated system of GPS-based navigation and object detection that will allow the visually impaired to reach any place in outdoor environments securely and effortlessly. Accurate recognition of the surrounding environment and real-time update of path planning are of paramount importance in such an outdoor navigation system to prevent various dangers. Combining the navigation algorithms with object recognition based on deep learning can enhance the contextual awareness of a system to better facilitate individual locomotion under complicated circumstances. And computationally efficient processing for real-time applications is crucial. It is a tool that helps people navigate outside in real time. The navigation safety is improved because the system is always checking the environment and changing to help people navigate safely with the system.

Algorithm: Real-Time Pathfinding, CNN (Deep Learning)

Merit: Enhanced navigational performance with on-time detection of surrounding obstacles that enhances autonomous navigation.

Demerit: Decreased performance in highly congested conditions depends on the computational ability of the device.

[2] TITLE: AN AI-BASED VISUAL NAVIGATION SYSTEM FOR THE BLIND (2023)

AUTHOR: Jordan Madden, Kolapo S. Alli, Lindon Falconer

In the current context, focus is on the advancement of intelligent visual navigation systems that aid in increasing mobility independence by employing both instantaneous object detection and the concept of a depth-aware navigation environment. Computer vision algorithms are integrated with mobile and/or wearable devices so that an individual is able to continuously observe their environment and dynamically be aware of obstacles around them. While designing such a system, emphasis is put on making the architecture lightweight, so it is easily portable and also comfortable to be worn by a user. Estimating the depth and identifying obstacles together provides efficient aid in traversability of paths with a view to safety for the user by recognising nearby obstacles and their distances. These intelligent assistive systems assist in navigating paths accurately while still providing the speed required in the outside world. This system is really important for giving people answers so they know what's going on in a world that is always changing. The system wants to make sure it works well no matter what's happening around it, so people can use it easily. This way of doing things is necessary for making a navigation system that actually helps people.

Algorithm: CNN (Convolutional Neural Network), Depth Estimation, Real-Time, Processing Models

Merit: Gives a real-time object detection along with the depth awareness for safe navigation.

Demerit: Hardware constraints, poor FPS (frame rate) and difficulties with dynamic or congested surroundings may result in poor detection.

[3] TITLE: A DEEP-LEARNING BASED MODEL TO ASSIST BLIND PEOPLE IN THEIR NAVIGATION (2022)

AUTHOR: Nitin Kumar, Dr Anuj Jain

A real-time human-aware system based on wearable smart devices and a smart cane based on deep learning is discussed in this paper for navigation of visually impaired people. The system comprises a vision-based wearable mask and a GPS-embedded smart stick, which will provide environmental information to the user in real time. A YOLOV5 trained with the combined datasets (i.e., MS COCO and the designed dataset for identifying navigation cues- left, right, straight) will be implemented using different sensors, including camera and ultrasonic, as input and a smart stick with a GPS device and Raspberry Pi as a processor for tracking the detected object. The accuracy recorded for the wearable model is 82.33%, whereas the smart stick gives an accuracy of 96.14%; the system has an overall accuracy of 89.24%. The developed system is a cost- effective approach with a detection rate of 45 FPS and more reliability in indoor and outdoor navigation. The navigation system is always ready. It keeps updating its decisions based on what's happening around the navigation system. This helps the navigation system respond well in situations. The navigation system uses what it sees and what the sensors tell the navigation system.

Algorithm: YOLOv5 (CNN-based Object Detection)

Merit: The system has an accurate, real-time object and path detection rate of 89.24%, making it a safe navigation device.

Demerit: The speed is highly dependent on the computing power of the hardware, and a larger training dataset may be needed to get a higher accuracy rate in more complex conditions.

[4] TITLE: AN OUTDOOR NAVIGATION ASSISTANCE SYSTEM FOR VISUALLY IMPAIRED PEOPLE IN PUBLIC TRANSPORTATION (2021)

AUTHOR: Salvador Martínez-Cruz, Luis A. Morales-Hernández, Gerardo I. Pérez-Soto, Juan P. Benítez-Rangel, Karla A. Camarillo-Gómez

The visually impaired users may not be able to individually operate the public transport system without appropriate identification for the bus and for the bus stop of the desired destination. The traditional GPS fails because of poor coverage of satellite signals and the use of an internet connection. However, a Bluetooth Low Energy (BLE) based framework can overcome this obstacle. This framework consists of installing beacon stations on each bus and bus stops so that the mobile phone receives information on its identity. This approach does not only guide to the bus and the destination stop but also keep the users updated about their position and the arrival to the bus stop using voice feedback from SUBE (a mobile application that recognizes beacons installed on buses and stops, and gives real time voice messages guiding to board and alight the bus or providing information of current position on the route). Tests on a real environment indicated more than 97% efficiency and better user independence, without the need for another wearable device.

Algorithm: BLE-based Signal Triangulation

Merit: Very precise and reliable aid on public transportation with the use of real-time BLE communication, which is not dependent on an uninterrupted satellite or internet connection.

Demerit: Requires dedicated BLE infrastructure installation and maintenance, making large-scale implementation complex and infrastructure-dependent.

[5] TITLE: AI BASED APP FOR BLIND PEOPLE (2021)

AUTHOR: Tufel Ali Qureshi, Mahima Rajbhar, Yukta Pisat, Prof. Vijay Bhosale.

An Android application using AI, this application is designed to help visually impaired persons in recognising objects and their environment with minimal to no external hardware. It captures live video streams from the mobile camera and processes the same using deep learning algorithms such as YOLO and Convolutional Neural Networks. The recognised object is then spoken out to the user using text-to-speech for effective real-time assistance in environment awareness. Text and currency recognition features are also implemented, which will greatly help users in their daily chores. The images undergo a certain amount of pre-processing, such as frame reduction and converting to grayscale, to speed up the recognition process. The results show a high accuracy for common objects and are also real-time. The system speaks out the recognised object to the visually impaired, allowing easier use of it. It only uses a smartphone. The application gives them feedback right away based on what it sees. The system is an easy way for visually impaired individuals to get around and know what is around them every day. It really helps with navigation and knowing what objects are, near visually impaired individuals.

Algorithm: YOLO, CNN (Convolutional Neural Network), TTS

Merit: Can effectively real-time object detection without using a smartphone, increasing independence for visually handicapped people.

Demerit: Its accuracy relies on the lighting and the camera's resolution, it consumes more battery if keep running.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

There are several technologies designed to assist navigation of visually impaired individuals, but none is a complete solution for a travel assistance system, which the supported systems mostly includes GPS-based navigation, object detectors, and transit announcement systems. Though they perform an auxiliary function for navigation, none is integrated as an intelligent system, but works alone separately.

A typical GPS-based system gives the direction and gives turn-by-turn instructions. But it was designed for normal sight. The information from GPS-based navigation could not recognise potential dangers in the environment, such as driving vehicles, roadblocks, ground differences, or obstacles. The GPS-based system requires good road conditions and does not guarantee that it can provide precise information during traffic.

There are a few obstacle detection assistive technologies that adopt ultrasound sensors or some wearable components, providing vibrational or audio prompts as an indication when detecting an object nearby. They only know that an object exists and could provide information about the distance of the obstacle, but failed to distinguish what the object is.

Recent advancements made using deep learning-based object detection models like YOLOv8 enable better performance to recognise obstacles in the environment via camera input, but most object detection-based system is work solely and are not connected with navigation systems and public transit. Transit announcements based on the bus tracking or beacon do not deal with environmental security.

Overall, there are currently no solutions that have adaptive intelligence and motion pattern analysis. It lacks features such as the ability to determine the class of an obstacle, to estimate the motion pattern of the user, and to automatically detect emergencies.

3.1.1 Limitations of Existing Systems

- Reduced system efficiency due to limited integration of Navigation, Obstacle detection, and transport assistance. Most of the presented solutions are independent systems and do not combine indispensable travelling characteristics into an intelligent system.
- Low awareness of environmental contexts leads to inefficient analysis of dynamically occurring risks. Real-time risks such as dynamic objects, obstacles and terrains are not recognised adequately.
- Extra hardware components make systems costlier and harder to maintain. These additionally equipped hardware components, like external sensors or wearable devices, limit mobility and extensibility.
- Poor recognition of the users' movement context-Current applications fail to distinguish if the user is travelling on foot, waiting for a bus or sitting in a bus.
- Reduced efficiency in crowded, noisy, dark environments- The reliability of GPS can decrease, and environmental variations bring about additional reduction in the performance of the system.
- The system does not change to fit how people use it and what they like. This means the system is not good, at helping people who move around in ways. They need kinds of navigation. The navigation system needs to adapt to map-based navigation. It also needs to process real-time data. The navigation system needs to work when the internet is not working well. It should help people navigate. The system should process data in time.

3.2 PROBLEM STATEMENT

It has been reported that visually impaired people find it difficult to travel through unknown paths and to use public transport by themselves. Existing aids offer limited services in the form of route finding or basic obstacle detection without merging all the important travel components within a smart device. The deficiency in intelligent real-time adaptive navigation, contextual awareness of obstacles, movement analysis and automatic handling of an emergency makes the travel of a visually impaired person riskier, and hence the person depends upon others for travelling. Therefore, an intelligent AI-based navigation aid with a merged system for path finding, obstacle detection, transport help, and verbal emergency support will help the visually impaired person to travel safely, securely, and independently.

3.3 PROPOSED SYSTEM

Smart Navigation and Travel Assistant An AI-driven integrated navigation and travel assistant is being designed to facilitate the mobility of a visually impaired person. The system is an entirely software-based Android and web-based application, which will not require any supplementary wearable device.

The process starts by securely registering a user by taking personal details and the number of a trustworthy person contactable in case of emergency. User interacts through speech only. Destination input can be given via speech, which will be recognised using the Speech Recognition technique. Real-time GPS position is obtained, and the optimum route is calculated using existing digital mapping applications such as OpenStreetMaps.

Constant tracking of the GPS position will keep the application dynamic throughout the journey. Using AI and Machine learning techniques, the type of motion by the user will be classified (Walking, Standing, Bus travel) using the inputs from the motion sensors and the GPS location to give an adaptive assistant.

While walking, the smartphone camera will be used to scan the surroundings for detecting any obstacles. Deep learning models like YOLOv8 can be used to detect real-time obstacles in images from the camera. This would include objects like vehicles, poles, steps, pedestrians and uneven roads, and alerts would be given using speech synthesis.

Distance estimation and priority-based warning algorithms are employed for timely and important obstacle announcement, while filtering algorithms prevent the system from giving repeated information thus increasing the clarity of warning for the users during motion. If it is determined to be traveling in a bus, the system keeps a check on the progression in the route and informs the users regarding the forthcoming bus stops. A warning is issued when a specific bus stop is reached to enable the users to board or de-board. Live location updates will ensure that users do not miss their bus stop or get into the wrong bus.

A dependable emergency handling feature has been incorporated. Pressing the screen for 5 long seconds initiates the SOS module; this module requires an initial confirmation within the first 3 seconds of touch on screen, by vibrating and giving voice cues, after which it sends an automatic emergency call along with location update and alert message to a pre-selected trusted contact.

Noise handling algorithms improve speech recognition in outdoor environments and reduce delay in obstacle alerts and navigation commands. The system also includes memory and power management to save battery. By combining GPS tracking, deep learning-based obstacle detection, motion detection, public transport assistance, and SOS support, it acts as a complete travel aid for visually impaired users. Its integration of perception, decision-making, and voice commands enhances safety, mobility, and confidence in both urban and semi-urban environments.

3.3.1 Advantages of Proposed System

- Integrated Navigation with obstacle detection and mobility assistance constitutes the holistic travelling support.
- All necessary modules are combined within one intelligent interface to provide smooth travel.
- AI-based dynamic environmental awareness provides real-time analysis of surroundings. Obstacles and changes are identified in real time for safe mobility.
- Classification of movements enables the system to recognise whether the user is walking, waiting or travelling by bus and to make context-aware decision-making.
- Intelligent incident detection with automatic SOS alerts enhances safety, and live location is shared with chosen contacts during incidents.
- A completely voice-controlled interface provides hands-free and vision-free operation; users provide voice inputs and receive feedback by means of voice.

CHAPTER 4

SYSTEM DESCRIPTION

4.1 SYSTEM SPECIFICATION

4.1.1 Hardware Requirements

- Processor : Octa-core / Multi-core smartphone processor
- RAM : Minimum 8 GB RAM
- Hard disk : 500 GB of storage
- Keyboard : Standard keyboard

4.1.2 Software Requirements

- Operating System : Windows 11, Linux
- Tools : VS Code, PyCharm
- Programming Language : Python 3.x
- Libraries : Numpy, pyttsx3, OSMnx, OpenCV
- Framework : Flask
- Version Control : GitHub API

Programming Language: Python

Python is an interpreted high-level programming language for general-purpose use, and it was created by Guido van Rossum and first released in 1991. Its design philosophy emphasises code readability with the notable use of significant white space, and it allows clear programming on both small and large scales. The leader of the language community, Guido Van Rossum, officially stepped down in July 2018, and Python is known to have a dynamic type system and automatic memory management, and it supports many programming paradigms, including object-oriented, imperative, functional and procedural, as well as a big and broad standard library. Interpreters are available for many platforms. His goal for the language was to create a small core with a large standard library and a readily extensible interpreter, and this came from his frustration with ABC, which had opposite plans. While being against "having too many choices for a method of programming", but opposed to extravagant syntax like that of Perl in favour of an uncluttered grammar.

Its creators intend to avoid "premature optimisation" and do not apply "patches to non-critical parts of Python" that might increase its speed slightly but decrease readability. They prefer to optimise performance in speed-critical code segments by moving the critical sections to extensions, which can be implemented in languages like C, or to use Pip, which is a JIT compiler, and it is available too, which would convert your Python script to C and perform native C API calls. Python programmers' target was keeping it "fun to program" and its name, a tribute to the British sketch comedy troupe Monty Python, along with tutorials/references which use amusing concepts like eggs and spam instead of for/bar common in technical contexts are all reflecting that goal, where on the contrary anything written messily and confusingly, that may appear as a transcription from another programming language is called "python".

CHAPTER 5

SYSTEM DESIGN

5.1 ARCHITECTURE DIAGRAM

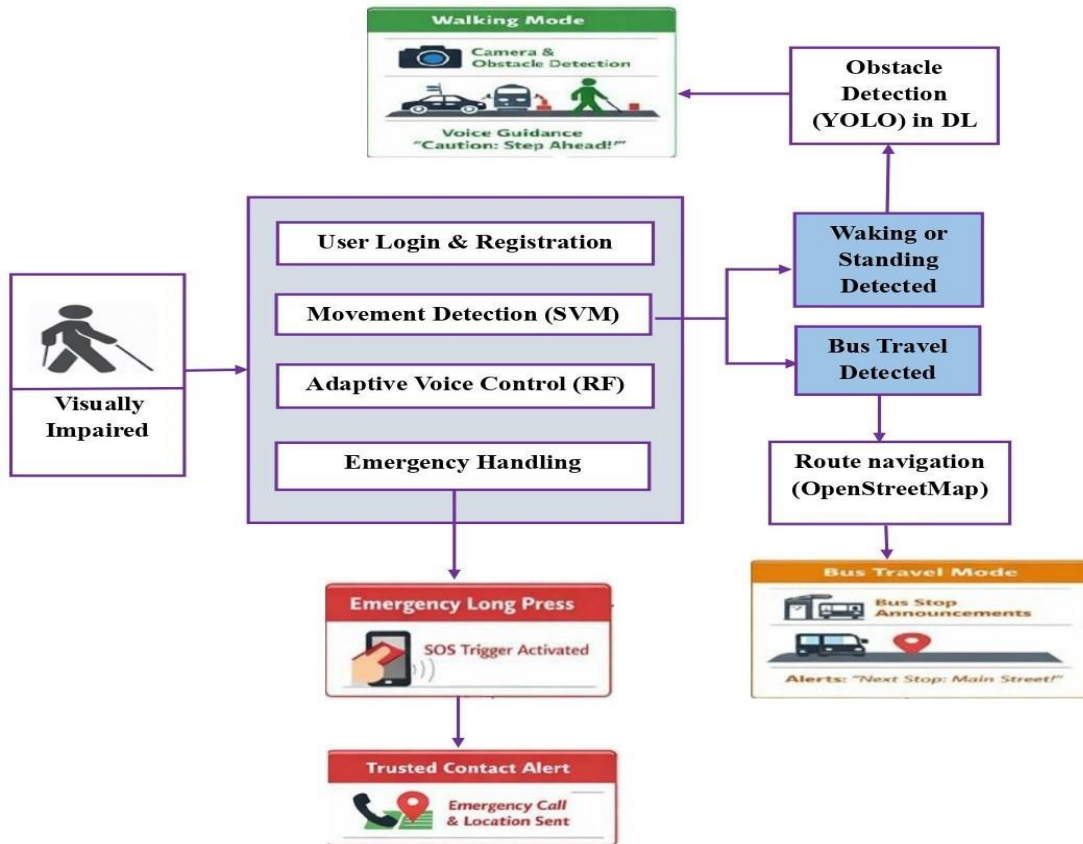


Figure: 5.1 System Architecture

1. User Input

The first step, the blind user will utilize his mobile application using his voice. The mobile application is assisted for secured authentication for the purpose of login, registration, user details and their trusted emergency contact. The user will then enter their desired destination after successful authentication via their voice. This whole operation makes a completely accessible input system.

2. Movement Detection (SVM)

Once the navigation has commenced, the sensor data from the mobile is continuously recorded. (GPS, accelerometer, gyroscope, etc.). The sensor readings from mobile are passed to the Support Vector Machine (SVM) model so as to classify the type of human activity (walking, standing, in a bus, etc.). This decision will make sure that the assistance will be dynamic and dependent on the present activity.

3. Walking Mode – Obstacle Detection (YOLO in Deep Learning)

Once walking or standing is detected, walking mode is activated. Real-time video from the camera of a smartphone is collected, and then analysed by the object detection of a deep learning model (YOLO). Obstacles such as vehicles, poles, steps, pedestrians, and uneven terrain are detected. According to distance and priority, the system will broadcast a warning to the user, for example, to warn about steps ahead or the proximity of obstacles. Safe navigation can be ensured in changing outdoor environments.

3. Bus Travel Detection and Route Navigation (OpenStreetMap)

Once bus travel is detected by speed and route information, the transport mode will be enabled. Using the user location information acquired by GPS and the path information provided by OpenStreetMap, the progress is monitored. When a user is approaching their destination stop or the next stop, corresponding warnings and announcements will be provided. The user can get on the right bus and safely step down at the right bus stop without help from others.

4. Adaptive Voice Control (RF)

The system adopted adaptive voice control methods. The speech recognition module converts speech into text, and the text-to-speech module delivers the navigational messages and warnings to the user. The use of noise filtering makes communication easier in public environments and outdoors.

5. Emergency Handling Mechanism

The emergency handling module is always running in the background. When long press screen for five seconds, it will trigger the SOS function. During the first three seconds, it will vibrate, and a voice confirms to enable cancellation. Once it is not cancelled, the message notification for the emergency alarm and call to a trusted person will occur automatically, and the user's live location will also be sent to the trusted person to accelerate the help during a critical time.

6. Trusted Contact Alert

When SOS is triggered, the trusted person will receive emergency notification with live tracking. This helps a person to have a quick response to provide help.

8. Continuous Monitoring and Feedback

It will be monitored for movement, position, and surrounding conditions all the time. Live processing with delay will achieve obstacle detection, route refresh and emergency alarm with minimum delay. Continuous feedback loop leads to a clever, reactive and secure trip guidance system.

5.2 DATAFLOW DIAGRAMS

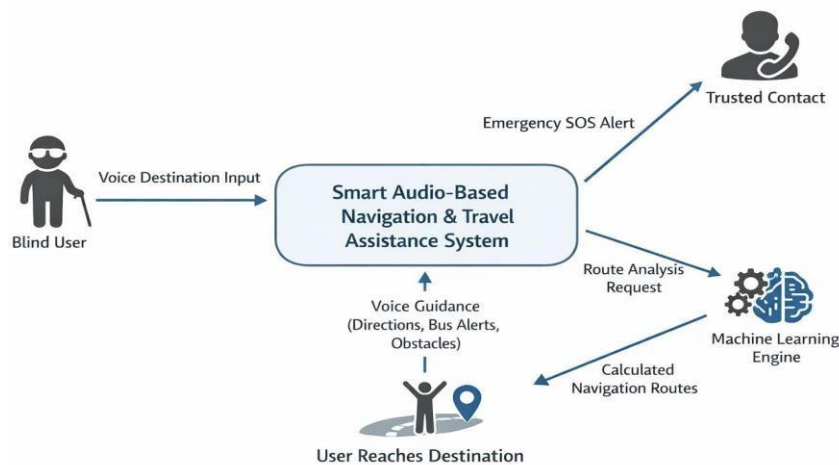


Figure: 5.2 Data Flow Diagram Level 0

This flow diagram shows the step-by-step procedure of the Smart Audio-Based Navigation and Travel Assistance System for blind people. The user inputs the destination using a voice command, the audio input is transformed into a request for route analysis, and this is transmitted to the machine learning engine. Using GPS and route information, the optimal path for navigation is analysed, and the analysed route is fed back to the main system. The user will then be assisted by continuous voice command, including directions, obstacles, and bus stop notifications. If the emergency occurred, an SOS message would be sent to the registered trusted contact, and the system would continue giving notice until the destination is achieved.

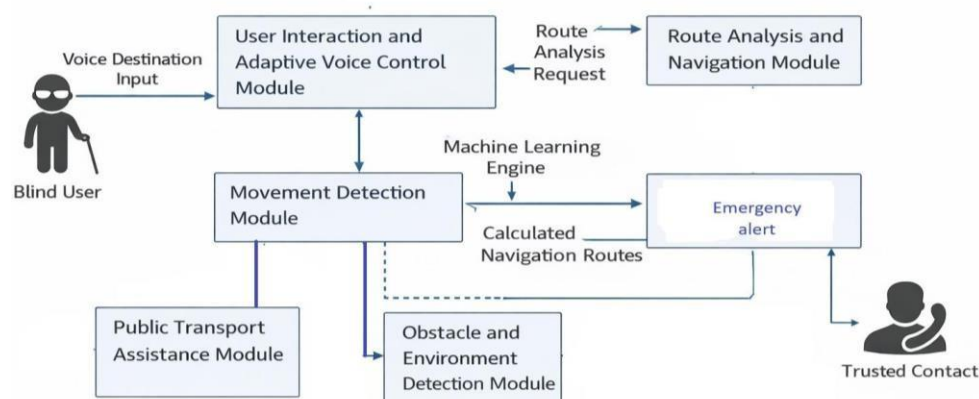


Figure: 5.3 Data Flow Diagram Level 1

The working of the system starts when the blind user inputs the destination orally via voice, which is further processed by the User Interaction and Adaptive Voice Control Module. The system sends the request to the Route Analysis and Navigation Module for the calculation of the best route with the help of the Machine Learning Engine. Simultaneously, it analyses the sensor input data by the Movement Detection Module to estimate the state of the user and subsequently activates the Public Transport Assistance Module for helping the user in their journey through buses or the Obstacle and Environment Detection Module for alerting during travel against obstacles and the navigation route calculated is given through continuous voice instructions. Finally, it will send an alert to the registered reliable person in case of any emergency.

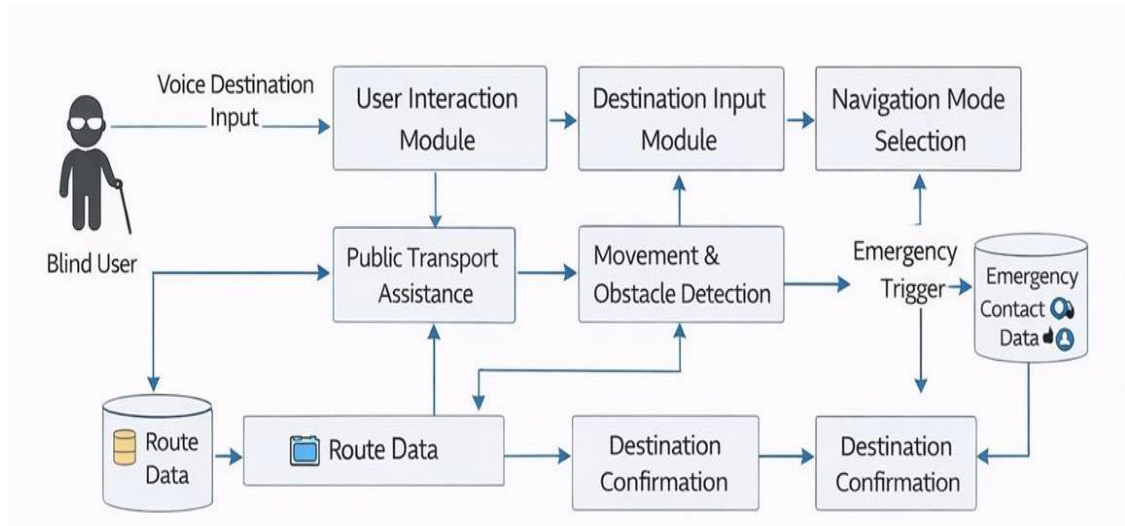


Figure: 5.4 Data Flow Diagram Level 2

This sequence begins when the user enters the desired destination using the voice input, and the destination is managed by the User Interaction Module to send to the Destination Input Module. By then, based on the request, the navigation mode to use is determined by the Navigation Mode Selection module. From the database, the corresponding route data are searched out and processed for navigation and public transport services. User activities and environment are monitored by the Movement and Obstacle Detection module to provide real-time alerts to the user. In case of an emergency situation, Emergency Trigger will use the saved Emergency Contact Data and sends an alert to the intended contacts. At the end, the destination is affirmed, and the journey is achieved with safe delivery.

The system continuously updates navigation instructions to handle route deviations effectively. Real-time synchronization between modules ensures smooth data flow and uninterrupted operation. The system is designed to handle dynamic environmental changes such as traffic and crowded areas. Efficient processing techniques are used to minimize delay and improve response time. The overall workflow ensures reliability, accuracy, and user safety throughout the journey.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 MODULES

A real-time AI-based navigation and obstacle detection system designed for visually impaired individuals includes multiple interconnected modules to ensure safe and independent mobility. The system begins with the "User Interaction and Voice Control Module," which allows the user to provide destination information through voice commands. Following this, the "Route Analysis and Navigation Module" calculates the most suitable route using GPS and digital maps. At the same time, the "Movement Detection Module" identifies whether the user is walking or using public transport. If the user is walking, the "Obstacle Detection Module" utilizes a camera and deep learning technology to recognize objects and potential hazards in the surroundings. In case of an emergency, the "Emergency SOS and Alert Module" activates to send alerts to the user's registered trusted contacts.

The following modules will be implemented to support real-time navigation and obstacle detection:

- User Interaction and Voice Control Module
- Route Analysis and Navigation Module
- Movement Detection Module
- Obstacle Detection Module
- Public Transport Assistance Module
- Emergency SOS and Alert Module

6.2 MODULES DESCRIPTION

6.2.1 User Interaction and Voice Control Module

This module serves as the interface between the visually impaired user and the navigation module. Voice commands are recorded using the smartphone microphone and fed to the speech recognition engine to translate them into text recognisable by the machine.

The destination is confirmed and feedback provided to the user in synthesised voice. Context-sensitive speech output will help the user during each step of the navigation to understand where and what to do. It is multilingual.

In noisy situations like street intersections, public transit stations or crowded markets, it uses voice command recognition through the noise cancellation, echo cancelling and adaptive audio processing algorithms, which can raise the precision to a very high level. It allows users to dictate commands even without touch.

6.2.2 Route Analysis and Navigation Module

The Route Analysis and Navigation Module plans the safest and fastest route from start to destination. It gets the live GPS coordinate and maps data to plan routes suitable for pedestrians. It involves a sophisticated routing logic that takes into account the safety of walking, the traffic management of the road crossing and the path continuation.

When there is a deviation of the actual path and the calculated path, it automatically recalculates the route without any delay. It keeps updating the directions and announces them using a constant audio navigation prompt. It estimates the travelling distance, the arrival time and the navigation progress. When there is a low GPS signal, and the device loses a signal, it updates the location at the right moment with live data.

6.2.3 Movement Detection Module

To recognise physical activities using sensor data from the smartphone, such as accelerometer, gyroscope, and GPS velocity. The sensor data are fused using sensor fusion to improve accuracy in movement recognition.

Then, features related to the physical activity, such as acceleration profile, step frequency, and speed, were used to feed into the machine learning algorithm. A classifier of SVM is applied to differentiate users' activity into 3 classes: walk, still, and bus riding. The system should also handle noisy or missing sensor data to ensure consistent activity recognition.

After identifying user activity, the system can perform context-aware assistance. The user's navigation behaviour would be adaptive to the user's activity and transition among assisting states seamlessly. The model can be further improved by training with diverse real-world datasets to enhance robustness.

6.2.4 Obstacle Detection Module

The obstacle detection module acts to give the pedestrian navigation system spatial perception of the environment. Through real-time analysis of camera inputs taken on the smartphone, the frames taken by the camera are processed by a deep learning model based on YOLO. The model is trained on images that simulate real-world obstacles. The system should also adapt to varying lighting conditions such as low light or bright sunlight for reliable detection

To optimise the efficiency of obstacle detection, the input frames are pre-processed by re-scaling them to a suitable size and normalising them. The model can detect objects such as cars, poles, stairs, walls, pedestrians, and bumps in the path. Algorithms can estimate distances to objects from the user. Critical hazards are addressed by making timely audio warnings so the pedestrian does not bump into obstacles. Low latency for real-time operation on a portable device is maintained by real-time optimization

6.2.5 Public Transport Assistance Module

This module guides the users while travelling in the bus. Based on the velocity & path taken determined by the Movement Detection Module, transport mode is initiated automatically.

The journey's status is determined using the current GPS position by comparing it with the saved path. The bus stops that are approaching are determined, and a voice alert for the stop is announced. The system should also handle unexpected route deviations such as bus diversions and update stop notifications accordingly.

The user is also notified of the bus stop which will be his destination before time, so that he can be prepared to get off. The route is constantly monitored to update the upcoming stops accurately, even during traffic delays. It can also provide estimated arrival time for upcoming stops to improve user preparedness.

6.2.6 Emergency SOS and Alert Module

Emergency SOS and Alert Module provides security in emergency cases. A signal search to verify if the user is interacting in any of the pre-defined emergency triggered modes, e.g., long press. This signal will be filtered to confirm and avoid accidental triggering using a countdown sequence.

The system should also allow multiple emergency contacts to be configured for increased safety. On verification of the emergency signal, the current location is updated, an emergency message with live GPS coordinates is sent to a designated trusted contact, and a phone call is initiated automatically.

The application keeps the monitoring active in the background, and during the process of navigation to the destination, the emergency service remains active. The module should ensure alert delivery confirmation and retry sending alerts if the initial attempt fails due to network issues.

ALGORITHM

Input: Voice commands, GPS coordinates, sensor inputs (accelerometer, gyroscope), camera video frames, emergency alert inputs.

Output: Navigation instructions, obstacle warnings, public transport updates, and emergency SOS alert.

Steps:

Step 1: Obtain Voice Command

The visually impaired person tells the voice recognition module the destination. The voice will be recorded.

Step 2: Speech to Text and feedback

Speech-to-text will convert the input voice into text, and the text-to-speech will confirm the input to the visually impaired person by speaking it out.

Step 3: Get GPS coordinates

The current location will be collected using the GPS sensor.

The path to be travelled by the visually impaired person will be fetched from the mapping database by finding out the safest possible route.

Step 4: Calculate the route and begin navigation

The shortest and safest possible path will be calculated, and the direction for the walk will be spoken to the visually impaired person, turn by turn.

Step 5: Get Sensor data

Continuous speed of the visually impaired person is obtained from accelerometer, gyroscope and GPS speed sensors. The activity can be classified by movement features like speed, acceleration, and motion type.

Step 6: Classify Movement

A supervised classifier, SVM, is used for classification between walking, stationary and bus activity. Automatic transition to walking, stationary or bus based on the detected features.

Step 7: Obstacle Detection (Walking)

The camera is enabled and will collect video frames in walking mode. The captured video frames will undergo pre-processing and then will be given as input to the YOLO for obstacle detection around the visually impaired person. Obstacle location, risk and range will be predicted.

Step 8: Public transport guidance (bus mode)

The location of the visually impaired person will be constantly tracked using GPS in the bus mode to track the journey. The closest bus stations will be stated.

Step 9: Navigation Guidance

Real-time guidance is provided by speaking turn-by-turn instructions. If an obstacle is detected or deviates from the current path, then a warning is given.

Step 10: Emergency Alert

Long-pressing the screen will trigger the alert. A counter is added to prevent false triggering.

Step 11: Activate SOS alert

If the emergency alert is activated, then the location of the visually impaired person will be sent to the designated trusted person, and a call will be made.

Step 12: Continuously monitor the system

The navigation path will be updated dynamically. The detection of obstacles and movement classification will be monitored until the destination is reached.

6.3 IMPLEMENTATION

The working of the Real Time AI Navigation and Obstacle Detection System for the Blind users consists of the following steps taken collectively. First, setting up the system and configuring the environment: this involves loading and setting up the GPS, Speech Recognition engine, camera and motion sensors in the Android system. Voice input acquisition: In this stage, the user has to feed the destination with his/her voice. The text input from the speech-to-text engine then verifies the destination and obtains real-time GPS coordinates for optimal route determination with the help of the map service.

Collecting the sensor data: at this stage, the sensor (Accelerometer, Gyroscope and GPS) readings are taken continuously. Features from the sensor data (speed, change in velocity, variance in motion) are extracted from the acquired data. This data is given as input to a machine learning model (SVM) for the identification of activity among walking, stationary or on-bus.

Activating the camera module: When the user is detected to be in "walk mode", the camera is activated, and frames are collected and processed in real-time. The frame size is reduced, and the data is normalised. The object detection model (YOLO) detects obstacles like cars, poles, steps, potholes, and human pedestrians; their distances are measured and accordingly alerted using text-to-speech (TTS).

Public Transport assistance module: In this module, GPS data is checked for movement patterns of a bus and compared to the data. Once bus stops are identified, the information about approaching stops is provided via text-to-speech.

Emergency SOS Module runs in the background continuously. When a specific predefined trigger is detected (such as a long-press gesture), the system shows a confirmation timer (to avoid wrong activation). Then, a real-time GPS location is obtained and sent to the trusted contact as an emergency notification message, and a calling notification is also displayed.

During the entire navigation process, it continuously updates the instructions to reroute if the user goes off the way. Processing of real-time data in a real-time environment is carried out using simple AI models, frame sampling and asynchronous calls, so as to reduce latency and power consumption.

The system incorporates various technologies:

- Android GPS and Location Services (for location tracking).
- Speech Recognition and Text to Speech (for voice input and output).
- Motion sensors (accelerometer, gyroscope for activity recognition).
- Camera API (for environmental awareness).
- Deep Learning frameworks (for obstacle recognition).
- SMS, call notification services (for emergency notification)

The sensor data and image data are processed locally to preserve the privacy of the user, and the system works independently of an internet connection. Testing of the system is performed under the real environment of streets and public transportation, focusing on the accuracy of navigation, precise obstacle detection, time response and stability of the system.

The system also emphasizes user safety and reliability by incorporating continuous monitoring and adaptive decision-making throughout its operation. It ensures that all modules function in coordination, allowing seamless interaction between navigation, obstacle detection, and emergency features without interruption. The design focuses on providing consistent performance even in uncertain or rapidly changing environments, thereby enhancing user confidence. By maintaining a balance between accuracy, speed, and resource efficiency, the system delivers a dependable and user-friendly experience for visually impaired individuals during real-time navigation.

The system is constantly being improved through new datasets and improved machine learning models, as well as efficient processing. Also, it is ensured that the navigation directions, obstacle alerts, and emergency notifications work accurately in a dynamic, real-time environment.

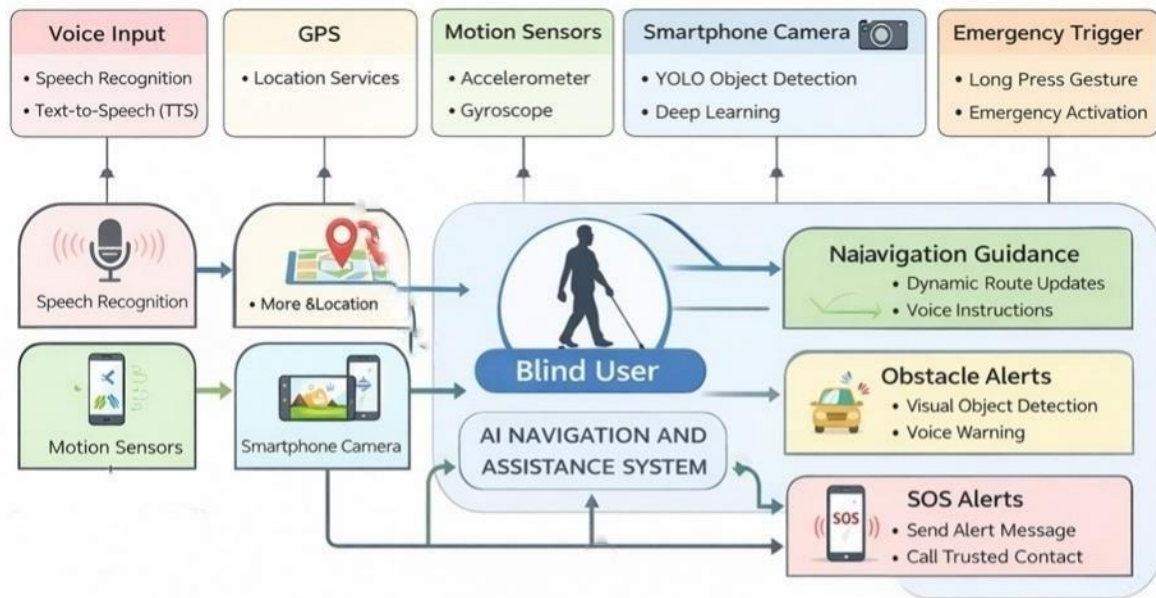


Figure 6.1: Implementation Diagram

The implementation diagram represents an AI-based navigation and assistance system designed for visually impaired users. The system integrates multiple inputs such as voice commands (speech recognition and text-to-speech), GPS for real-time location tracking, motion sensors (accelerometer and gyroscope) for movement detection, and a smartphone camera using YOLO-based deep learning for object detection. The obstacle detection system can handle dynamic and moving objects in complex environments. These components work together to provide navigation guidance with dynamic route updates and voice instructions. The system also detects obstacles and alerts the user through voice warnings. The system is designed to work efficiently under varying environmental conditions such as low light and crowded areas. Additionally, an emergency feature is included, where a long-press gesture triggers SOS alerts by sending messages or calling trusted contacts. The system supports scalability for integrating additional features like indoor navigation in the future. Overall, the system combines perception, decision-making, and voice interaction to ensure safe and independent navigation.

CHAPTER 7

SYSTEM TESTING

7.1 TESTING OBJECTIVES

System Testing is performed to verify that the Real-Time AI Navigation and obstacle Detection System is working accurately, reliably, and safely in the actual, real-life conditions. The purpose of this test is to ensure that all modules work as expected when working alone as well as in the whole system.

Objectives of this test are:

- Verify correct speech recognition and voice output.
- Verify correct route calculation and updating of the route in real time.
- Verify movement classification (walking, standing, and travelling on a bus).
- Verify real-time obstacle detection and appropriate voice alerts.
- Verify the correct public transport stop announced.
- Verify that the Emergency SOS module works as expected.
- Test the latency and system reaction time.
- Verify the overall reliability outdoors.

Testing is performed in both controlled and real-world conditions, including streets, intersections, and public transportation environments.

7.2 TYPES OF TESTING

Unit Testing

Individual modules (Speech recognition, GPS tracking, movement classification, Obstacle detection and SOS alert) are tested in unit tests to determine whether the module behaves as specified.

Integration Testing

Integration testing examines how individual modules communicate with each other. (e.g., after movement classification, determine that walking mode is detected, the obstacle detection module is activated). The voice guidance with navigation was also tested in this stage.

System Testing

System testing measures the entire application in its totality as a system. It tests the entire application from the system when it is given a destination till the final destination is reached to find bugs and ensure its seamless functioning.

Real-Time Testing

Because it is a real-time system, we test its performance under the changing real-world conditions such as busy roads, moving vehicles, and different light conditions. The system should maintain stable performance even during rapid changes in environmental conditions.

Performance Testing

The performance tests will evaluate the

- Obstacle detection lag
- Voice response lag
- Accuracy of the GPS track
- Battery consumption
- continuous system load performance of the system
- System response consistency under prolonged continuous usage

User Acceptance Testing

The system is tested on a sample of users to verify the user-friendliness, the intelligibility of the instructions given by voice, and the user satisfaction with the system, and to identify aspects for improving the speed and usability of the system. The system should collect user feedback to further enhance usability and accessibility features.

7.3 TEST CASE

Voice Recognition and Preprocessing Test Cases

Testing the system on the noisy voice inputs and variations of speech is crucial during pre-processing. It should recognise voice commands on the destination, and it must be correctly translated into text from the speech recognition module without any errors outdoors, even in the presence of noise. The system should again request the voice in case of unclear audio input; noise filtering is to be used, so that the system does not fail/ misinterpret the voice commands. The validated destination from the recognised speech should be confirmed before navigation to that place begins.

Navigation Accuracy Test Cases

A GPS navigation module to be verified on the correctness of position and route calculation, and also dynamic re-routing of the route for correct navigation. It should also compute a route that will be the safest and minimise walk time for the pedestrians. In the case of deviation from the allocated route, the module should realise the deviation and immediately re-route with updated voice prompts. GPS drift and signal loss due to environmental factors should be tolerated. The system should provide continuous voice guidance updates at regular intervals to keep the user informed throughout navigation.

Movement Classification Accuracy Test Cases

Metrics such as accuracy need to be applied to the measured accelerometer and gyroscope data, along with GPS speed, in order to verify the implemented model for movement detection. The model is expected to detect the transportation mode from walking, standing and bus transportation. Measures such as accuracy, precision and recall should be used during validation. For non-standard conditions such as slow walk, rapid stop and start or corrupted sensor readings, there must be no misclassification of modes. In addition, the system must be able to seamlessly switch between the states automatically. The system should maintain consistent performance even during sudden environmental changes or irregular user movements

Obstacle Detection and Alert Test Cases

The deep learning based obstacle detection module should be evaluated, verifying the detection accuracy, reaction time, and reliability of the warning. Car, pole, stairs, human, etc. Should be accurately detected by the YOLO model. When the obstacle is too close to the person, a voice warning should be provided. It should also show good performance in different lighting situations and crowded areas. The system should also be tested for detecting partially visible or moving obstacles to ensure consistent performance in dynamic environments.

Public Transport Assistance Test Cases

The progress of the route is expected to be recognised, and the movement of the bus should be identified by the route assistance module. Bus routes should be detected accurately, and the notifications should be sent promptly before the bus reaches the destination stop. There should be no spurious stop notifications because of GPS errors or wrong route tracing. Voice notifications should be tested to check for clear and on-time announcements so that passengers get enough time to enter or exit. The system should also handle sudden route changes or delays in public transport and update notifications accordingly.

Emergency SOS Functionality Test Cases

The emergency module has to be tested to ensure it triggers reliably via a long-press gesture. The confirmation timer must avoid unintentional triggers. When a long press has been confirmed, it has to get the location accurately and send an alert SMS and a phone call to the emergency contact with minimal latency.

End-to-End Workflow Test Cases

The whole system workflow should be evaluated to realize whole cycle works from the destination input until reaching the end destination. Real-time sensor input reading should be sustained. Obstacle detection accuracy and real-time route update should work. An emergency monitoring system should be robust, non-crashing, system performance over the long run.

System Integration and Edge Case Test Cases

The integrated system should be able to handle real-time streams of data coming from several modules simultaneously. It should be able to not crash when subjected to adverse situations such as very low light, GPS loss, noise from the environment, a crowd of people, spontaneous motion by the user, etc. It should be robust enough for processing the entire stream of camera as well as sensor data.

Test Case for Classification Input (Movement Detection Model)

Input:

- Sensor input data(Accelerometer readings, gyroscope data)
- GPS speed input
- Real-time motion signatures
- Variation in environmental motion

Expected Output:

- Classification output (Walking/Standing/Bus movement)
- Confidence probability score (if present)

To develop a test case for classification input, the sensor data input needs to be preprocessed by removing noise and extracting useful features from the data (e.g., speed, variation of acceleration and frequency of motion). The trained ML model detects the patterns of motion and classifies them. The edge cases that need to be included are the test cases of slow walking, stopping of the vehicle, road imperfections, and signal interference. High classification accuracy and automated detection of the appropriate modules, such as obstacle detection and transportation support, should be maintained. Additionally, the system should be tested for handling incomplete or delayed input data without affecting overall performance.

The comprehensive set of test cases developed above would verify the entire AI navigation system, right from the voice input to the obstacle detection, transport support and alert message generation. The system should also ensure scalability and consistent performance when multiple modules operate simultaneously under heavy load conditions.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 PERFORMANCE ANALYSIS

Performance of the Real-Time AI Navigation and Obstacle Detection System for Blind Users: A thorough investigation of performance was undertaken for the following parameters. These include the system's accuracy in recognising and detecting various factors, response rate to navigation and alerts, reliability of navigation and stability of the system. Our system uses machine learning for the movement detection module and deep learning for obstacle detection. These are integrated to provide support in real-time.

Movement Detection Module: The SVM-based movement detection module was found to have an average classification accuracy in the range of 90-93% in distinguishing among modes like walking, standing and travelling in a bus. In many cases, the model identified the mode accurately even with slight differences in motion and velocity in testing conditions.

Obstacle Detection Module: The YOLO based obstacle detection module gave promising detection reliability for common obstacles expected in an outdoor environment, such as vehicles, poles, stairs, people, etc. The processing speed gave us a real-time response, which ensured low latency to produce the voice feedback to the user.

Navigation Module: The navigation module utilizing GPS, gives accurate results on path guidance and route deviation error. Even if there was a deviation in the route, the route was recalculated and presented within 2-3 seconds, and a correct voice message was displayed and announced to the user.

Emergency SOS Module: The Emergency SOS module responded at once once the long press was confirmed, and a location detail was sent to the predefined contact in no time. Overall, the system showed stable real-time performance in an outdoor environment and helped a visually impaired person to a great extent.

Accuracy

For the performance evaluation of the Movement Detection Model, we use accuracy as the metric. Accuracy defines the percentage of activities correctly detected out of all the detected activities of the user. This metric provides a clear understanding of how well the system identifies user movements under different conditions. Accuracy is calculated as follows:

$$\text{Accuracy} = \text{No. of correct predictions} / \text{No. of total predictions}$$

To further validate the model's performance, use a confusion matrix for performance evaluation.

Confusion Matrix for Movement Classification

Predicted Values	Actual Walking	Actual Standing	Actual Bus
Walking	TP	FP	FP
Standing	FN	TP	FP
Bus	FN	FN	TP

Where:

TP – True Positive (Correctly classified activity)

TN – True Negative

FP – False Positive

FN – False Negative

$$\text{Total Instances} = \text{TP} + \text{TN} + \text{FP} + \text{FN}$$

Using experimental test data, the system achieved approximately 92% classification accuracy.

Navigation Accuracy

Accuracy in navigation deals with how accurately the system directs the user towards the intended destination. Location accuracy when the user was moving, like walking, was well within expected limits. Route recalculation speed was prompt when the user deviated.

Response Time

Response time is of high importance in any real-time system, especially in assistive technologies where delays can affect user safety. Performance tests recorded the following times for various functions:

- Voice recognition response time: Rapid and smooth, enabling quick command processing
- Obstacle detection time delay: Near real-time, allowing immediate alerts
- Route recalculation time: A few seconds, ensuring uninterrupted navigation
- SOS alert transmit time: Instant when confirmation is received

The fast response time has been proven useful in ensuring the safety of users while they navigate, particularly in dynamic environments.

System Stability

The system was tested over an extended period under continuous usage conditions. Camera stream processing, sensor data input, and GPS data processing were handled simultaneously without any system crashes or performance degradation. Efficient memory management and optimization techniques ensured smooth operation. Optimization has also improved battery life, making the system suitable for long-duration use without frequent recharging.

Overall Discussion

The above results have proven the AI Navigation and Obstacle Detection System's reliability in real-world conditions. The high accuracy achieved in activity classification, combined with real-time obstacle detection and stable navigation performance, demonstrates the effectiveness of the system. Additionally, the quick response to emergency signals further enhances user safety. These results provide strong evidence of how beneficial the system can be. The system has proved its potential to encourage independence and safety for visually impaired individuals while navigating their surroundings.

CHAPTER 9

CONCLUSION AND SCOPE FOR FUTURE WORK

9.1 CONCLUSION

The presented real-time artificial intelligence navigation and obstacle detection system overcomes the mobility issue of visually impaired persons effectively. This system integrates machine learning, deep learning, GPS navigation, and real-time voice-based guidance and helps the blind navigate the outdoor environment safely. The Movement Detection Module correctly identifies different user movements, namely walking, standing, travelling on a bus, etc., with greater accuracy. The YOLO based Obstacle Detection Module helps in detecting and identifying nearby obstacles in real time and immediately gives a voice-based warning so as to avoid any mishap. The navigation module not only guides on the routes with reliable accuracy and dynamically changes the route if deviated, but also the public transport assistant helps them in easily travelling alone by informing them of bus stops with good accuracy and at the right time. The emergency SOS module guarantees their safety in an emergency and helps them immediately. In conclusion, this system leads to more independence for blind persons, increases their safety and reduces their dependence on people. Experiments also show that the devised method is accurate, efficient and ready for deployment.

9.2 SCOPE FOR FUTURE WORK

The proposed system can be further extended by: increasing the accuracy of obstacle detection under difficult weather conditions and low illumination using deep learning networks, advanced algorithms and fusion with other sensors.

Incorporating LiDAR or ultrasonic sensors can increase the accuracy of depth perception and distance measurement. Integrating with Indoor Navigation using Bluetooth beacon technology or Wi-Fi-based indoor positioning systems, multilingual support, and user personalised preferences. Integration with the smart city systems and the public transportation APIs to provide real-time traffic and bus tracing information.

We will also investigate wearable device implementation (like smart glasses or other lightweight modules) and adaptive learning techniques (to improve performance according to the user's movement pattern over time). Battery optimisation and computational efficiency need further research for the scalability and the effectiveness for continuous operation.

APPENDICES

APPENDIX 1

SOURCE CODE

Obstacle detection.py

```
import time
import threading
import json
import queue
import numpy as np
import pyttsx3
from flask import Flask, request, jsonify, send_from_directory, Response,
stream_with_context
from collections import deque
import cv2
from ultralytics import YOLO
import os
import socket
app = Flask(__name__)
# TEXT TO SPEECH
tts_engine = pyttsx3.init()
tts_engine.setProperty('rate', 155)
tts_engine.setProperty('volume', 1.0)
voices = tts_engine.getProperty('voices')
for v in voices:
    if 'zira' in v.name.lower() or 'david' in v.name.lower():
        tts_engine.setProperty('voice', v.id)
break
tts_lock = threading.Lock()
_speech_queue = deque()
_speech_queue_lock = threading.Lock()
_speech_event = threading.Event()
def _enqueue(text, priority=False):
    with _speech_queue_lock:
        if text in _speech_queue:
            return
        if priority:
```

```

_speech_queue.appendleft(text)
else:
    speech_queue.append(text)
speech_event.set()
def _tts_worker():
    while True:
        _speech_event.wait()
        while True:
            with _speech_queue_lock:
                if not _speech_queue:
                    _speech_event.clear()
                    break
            text = _speech_queue.popleft()
            with tts_lock:
                try:
                    tts_engine.say(text)
                    tts_engine.runAndWait()
                except Exception:
                    pass
            threading.Thread(target=_tts_worker, daemon=True).start()
def speak_movement(text):
    broadcast_sse("speak", {"text": text})
    _enqueue(text, priority=False)
def speak_obstacle(text):
    broadcast_sse("speak_urgent", {"text": text})
    _enqueue(text, priority=True)
def speak_force(text):
    broadcast_sse("speak_urgent", {"text": text})
    _enqueue(text, priority=True)
# OBSTACLE TRACKER — max 2 announcements, then periodic
RE_ANNOUNCE_SEC = 8.0
MAX_FIRST_BURST = 2
class ObstacleTracker:
    def __init__(self):
        self.lock = threading.Lock()
        self.current_text = ""
        self.announce_count = 0
        self.last_said_time = 0.0

```

```

def should_speak(self, text):
    with self.lock:
        now = time.time()
        if text != self.current_text:
            self.current_text = text
            self.announce_count = 1
            self.last_said_time = now
            return True
        if self.announce_count < MAX_FIRST_BURST:
            if (now - self.last_said_time) >= 1.5:
                self.announce_count += 1
                self.last_said_time = now
            return True
        return False
    if (now - self.last_said_time) >= RE_ANNOUNCE_SEC:
        self.last_said_time = now
        return True
    return False

def clear(self):
    with self.lock:
        self.current_text = ""
        self.announce_count = 0
        obs_tracker = ObstacleTracker()
        # MOVEMENT THRESHOLDS
        STILL_MAX = 0.04
        STANDING_MAX = 0.40
        WALKING_PEAKS_MIN = 3
        PEAK_THRESHOLD_OFFSET = 0.20
        MIN_PEAK_GAP = 6
        IDLE_TIMEOUT_SEC = 8.0
        MAJORITY_THRESHOLD = 0.6
        CONFIRM_WINDOW = 5
        # CAMERA THREAD
        model = YOLO("yolov8n.pt")
        camera_active = False
        camera_thread = None
        camera_lock = threading.Lock()
        clear_path_counter = 0

```

```

last_clear_announce = 0.0
CLEAR_PATH_FRAMES = 8
CLEAR_ANNOUNCE_GAP = 12.0
def obstacle_detection_loop():
    global camera_active, clear_path_counter, last_clear_announce
    cap = cv2.VideoCapture(0)
    if not cap.isOpened():
        print("✖ Camera not opened")
        camera_active = False
        with obstacle_lock:
            obstacle_state["camera_active"] = False
        return
    cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
    cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)
    cap.set(cv2.CAP_PROP_FPS, 15)
    print("📷 Obstacle detection started")
    obs_tracker.clear()
    clear_path_counter = 0
    with obstacle_lock:
        obstacle_state["camera_active"] = True
        obstacle_state["last_obstacle"] = ""
        obstacle_state["last_urgency"] = 0
    while camera_active:
        ret, frame = cap.read()
        if not ret:
            time.sleep(0.5)
            continue
        results = model(frame, verbose=False)
        if not camera_active:
            break
        h, w, _ = frame.shape
        detections = []
        for r in results:
            for box in r.bboxes:
                conf = float(box.conf[0])
                class_id = int(box.cls[0])
                if conf < 0.45: continue
                x1, y1, x2, y2 = map(int, box.xyxy[0])

```

```

box_h  = y2 - y1
obj_cx  = (x1 + x2) // 2
distance = estimate_distance(box_h, class_id, h)
position = horizontal_zone(obj_cx, w)
_, lbl  = OBJECT_REF.get(class_id, (DEFAULT_REF_HEIGHT, "obstacle"))
if distance <= ZONE_WARN:
    detections.append((class_id, distance, position, lbl))
if detections:
    clear_path_counter = 0
    best = pick_most_dangerous(detections)
    if best:
        cls_id, dist, pos, lbl = best
        instruction, urgency  = build_instruction(lbl, pos, dist)
        # Update dashboard state always (even if suppressed in TTS)
        with obstacle_lock:
            obstacle_state["last_obstacle"] = instruction
            obstacle_state["last_urgency"] = urgency
            obstacle_state["last_update"]  = time.time()
            if obs_tracker.should_speak(instruction):
                speak_obstacle(instruction)
                print(f"🚧 [{urgency}★] {instruction}")
            else:
                print(f"🔇 [{urgency}★] suppressed: {instruction}")
            else:
                obs_tracker.clear()
                clear_path_counter += 1
                # Update dashboard — clear obstacle
                with obstacle_lock:
                    obstacle_state["last_obstacle"] = ""
                    obstacle_state["last_urgency"] = 0
                now = time.time()
                if (clear_path_counter >= CLEAR_PATH_FRAMES and
                    (now - last_clear_announce) > CLEAR_ANNOUNCE_GAP):
                    speak_obstacle("Path is clear. Continue walking.")
                    last_clear_announce = now
                    clear_path_counter = 0
                print(f"✅ Path clear")
                with obstacle_lock:

```

```

obstacle_state["last_obstacle"] = "Path is clear"
obstacle_state["last_urgency"] = 0
time.sleep(0.35)
cap.release()
obs_tracker.clear()
with obstacle_lock:
obstacle_state["camera_active"] = False
obstacle_state["last_obstacle"] = ""
obstacle_state["last_urgency"] = 0
print("📷 Obstacle detection stopped")
def manage_camera(final_state):
    """Update camera state for frontend to react to"""
    global camera_active
    with camera_lock:
        if final_state == "WALKING" and not camera_active:
            camera_active = True
            with obstacle_lock:
                obstacle_state["camera_active"] = True
            print("📷 Camera should be active (frontend)")
        elif final_state != "WALKING" and camera_active:
            camera_active = False
            with obstacle_lock:
                obstacle_state["camera_active"] = False
            print("📷 Camera should be inactive (frontend)")
    # Get the image data from request
    frame_data = request.get_data()
    if not frame_data:
        return jsonify({"error": "No frame data"}), 400
    # Convert bytes to numpy array
    nparr = np.frombuffer(frame_data, np.uint8)
    frame = cv2.imdecode(nparr, cv2.IMREAD_COLOR)
    if frame is None:
        return jsonify({"error": "Invalid image"}), 400
    # Process frame with YOLO
    results = model(frame, verbose=False, conf=0.30) # Lower confidence threshold
    to 0.30 for better detection
    h, w, _ = frame.shape
    detections = []

```

```

total_detections = 0
for r in results:
    for box in r.bboxes:
        total_detections += 1
        conf = float(box.conf[0])
        class_id = int(box.cls[0])
        # Log all detections for debugging
        class_name = OBJECT_REF.get(class_id, (None, "unknown"))[1]
        print(f" Detected: {class_name} (ID:{class_id}) - Confidence: {conf:.2f}")
        if conf < 0.30: # Skip very low confidence detections
            continue
        x1, y1, x2, y2 = map(int, box.xyxy[0])
        box_h = y2 - y1
        obj_cx = (x1 + x2) // 2
        distance = estimate_distance(box_h, class_id, h)
        position = horizontal_zone(obj_cx, w)
        _, lbl = OBJECT_REF.get(class_id, (DEFAULT_REF_HEIGHT, "obstacle"))
        if distance <= ZONE_WARN:
            detections.append((class_id, distance, position, lbl, conf))
            print(f" ⚠ {lbl} at {distance}m, {position} (conf: {conf:.2f})" if
            detections:
            # Update obstacle state
            with obstacle_lock:
                obstacle_state["last_obstacle"] = instruction
                obstacle_state["last_urgency"] = urgency
                obstacle_state["last_update"] = time.time()
            # Speak if needed
            if obs_tracker.should_speak(instruction):
                speak_obstacle(instruction)
                print(f" 🗣 [{urgency} ★] {instruction}")
            return jsonify({
                "status": "ok",
                "obstacle": {
                    "text": instruction,
                    "urgency": urgency
                }
            })
        else:

```

```

obs_tracker.clear()
with obstacle_lock:
    obstacle_state["last_obstacle"] = ""
    obstacle_state["last_urgency"] = 0
    if total_detections > 0:
        print(f"      i   Detected   {total_detections}   object(s)   but   none   within
        {ZONE_WARN}m")
    else:
        print(f" ✓ No obstacles detected")
    return jsonify({
        "status": "ok",
        "obstacle": {
            "text": "Path clear. No obstacles.",
            "urgency": 0
        }
    })
except Exception as e:
    print(f" ✖ Error processing camera frame: {e}")
import traceback
traceback.print_exc()
return jsonify({"error": str(e)}), 500
@app.route("/camera/start", methods=["POST"])
def camera_start():
    return jsonify({"status": "ok", "message": "Camera ready"})
def _contacts_candidates():
    script_dir = os.path.dirname(os.path.abspath(__file__))
    return [
        os.path.join(script_dir, "contacts.json"),
        os.path.join(os.path.dirname(script_dir), "contacts.json"),
    ]
def load_trusted_contacts():
    for path in _contacts_candidates():
        if not os.path.exists(path):
            continue
        try:
            with open(path, "r", encoding="utf-8") as f:
                raw = json.load(f)
            contacts = []

```

```

if isinstance(raw, dict):
for name, phone in raw.items():
contacts.append({"name": str(name), "phone": str(phone)})
elif isinstance(raw, list):
for item in raw:
if isinstance(item, dict):
name = str(item.get("name", "Unknown"))
phone = str(item.get("phone", ""))
if phone:
contacts.append({"name": name, "phone": phone})
return contacts
except Exception as err:
print(f"⚠ Failed to load contacts from {path}: {err}") return []
@app.route("/sos/contacts", methods=["GET"])
def sos_contacts():
contacts = load_trusted_contacts()
return jsonify({"count": len(contacts), "contacts": contacts})
# MAIN
if __name__ == "__main__":
speak_force("Navigation system ready. Please start moving.")
# Get IP address for phone access
print("📶 Blind Navigation Backend — port 5000")
print("-" * 60)
print(f"🧘 SITTING → mag_std < {STILL_MAX}")
print(f"🚶 WALKING → mag_std > {STANDING_MAX} AND peaks >= {WALKING_PEAKS_MIN}")
print("-" * 60)
print(f"🛑 CRITICAL : ≤ {ZONE_CRITICAL} m → STOP immediately")
print(f"🚧 CLOSE : ≤ {ZONE_CLOSE} m → slow down / turn")
print(f"⚠ WARN : ≤ {ZONE_WARN} m → heads-up")
print("-" * 60)
print(f"📱 Open on your PHONE:")
print(f" {protocol.upper()}: {protocol}://localhost:5000 (if on same device)")
if ip_address != "YOUR_IP":
print(f" OR: {protocol}://{ip_address}:5000")
print(f"💻 Open on LAPTOP:")

```

```

print(f" {protocol}://127.0.0.1:5000")
if protocol == "https":
    print(f"\n⚠ If you see 'Not Secure' warning:")
    print(f" Click 'Advanced' → 'Proceed' (it's your own certificate)")
else:
    print(f"\n⚠ Running on HTTP - mobile sensors may not work!")
    print(f" HTTPS certificates are required for GPS and DeviceMotion APIs.")
    print("\n" + "-" * 60 + "\n")
# Start server
if os.path.exists(cert_file) and os.path.exists(key_file):
    print("🔒 Starting HTTPS server...\n")
    app.run(host="0.0.0.0", port=5000, ssl_context=(cert_file, key_file),
            threaded=True)
else:
    print("⚠ Starting HTTP server (limited mobile support)...\n") app.run(host="0.0.0.0",
    port=5000, threaded=True)

```

Navigation.py

```

from flask import Flask, request, jsonify, send_from_directory
import threading, time
import osmnx as ox
import networkx as nx
import speech_recognition as sr
import pyttsx3
from geopy.geocoders import Nominatim
from geopy.exc import GeocoderServiceError
from geopy.distance import geodesic
import numpy as np
import os
import json
import logging
app = Flask(__name__)
current_location = None
destination = None
language_mode = "english"
# Set directories at module load time (before any Flask request context)

```

```

SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
PROJECT_DIR = os.path.dirname(SCRIPT_DIR)
FRONTEND_DIR = os.path.join(PROJECT_DIR, "frontend")
LOCATIONS_FILE = os.path.join(SCRIPT_DIR, "locations.json")
logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)
@app.after_request
def add_cors_headers(response):
    response.headers["Access-Control-Allow-Origin"] = "*"
    response.headers["Access-Control-Allow-Headers"] = "Content-Type, Authorization"
    response.headers["Access-Control-Allow-Methods"] = "GET, POST, PUT, DELETE, OPTIONS"
    return response
@app.route("/frontend/<path:filename>")
def serve_frontend(filename):
    return send_from_directory(FRONTEND_DIR, filename)
def load_local_locations():
    if not os.path.exists(LOCATIONS_FILE):
        return {}
    try:
        with open(LOCATIONS_FILE, "r", encoding="utf-8") as file:
            data = json.load(file)
            if isinstance(data, dict):
                return data
            return {}
    except Exception as error:
        logger.error(f"Failed to load locations.json: {error}")
        return {}
def find_location(text, local_locations):
    query = (text or "").strip().lower()
    if not query:
        return None
    for location_name, coordinates in local_locations.items():
        if location_name.lower() == query and isinstance(coordinates, (list, tuple)) and len(coordinates) >= 2:
            return location_name, (coordinates[0], coordinates[1])

```

```

for location_name, coordinates in local_locations.items():
    lower_name = location_name.lower()
    if (query in lower_name or lower_name in query) and isinstance(coordinates, (list,
tuple)) and len(coordinates) >= 2:
        return location_name, (coordinates[0], coordinates[1])
    return None

engine = pyttsx3.init()
engine.setProperty("rate", 150)
tts_lock = threading.Lock()
gps_announced = False

def speak(text):
    try:
        print("🔊", text)
        with tts_lock:
            engine.say(text)
            engine.runAndWait()
    except Exception as error:
        logger.warning(f"TTS error: {error}")

def listen():
    r = sr.Recognizer()
    with sr.Microphone() as source:
        r.adjust_for_ambient_noise(source, duration=1)
    try:
        audio = r.listen(source, timeout=7, phrase_time_limit=7)
    except:
        return None
    try:
        text = r.recognize_google(audio)
        print("🗣️ You said:", text)
        return text.lower()
    except:
        return None

@app.route("/")
def home():
    return """
<html>
<body>

```

```

<h2>Blind Navigation</h2>
<button onclick="sendGPS()">Send GPS</button>
<script>
function sendGPS() {
navigator.geolocation.getCurrentPosition(function(pos) {
fetch("/location", {
method: "POST",
headers: {"Content-Type": "application/json"},
body: JSON.stringify({
lat: pos.coords.latitude,
lon: pos.coords.longitude
})
});
alert("GPS Sent Successfully");
});
}
</script>
</body>
</html>
"""Return the current GPS location"""
global current_location
if current_location is None:
# Return default Chennai coordinates if no GPS location yet
return jsonify({
"status": "no_location",
"lat": 13.0827,
"lon": 80.2707,
"message": "No GPS location received yet"
})
return jsonify({
"status": "success",
"lat": current_location[0],
"lon": current_location[1]
})
def select_language():
global language_mode
speak("Say Language: Tamil or English")

```

```

lang = listen()
if lang is None:
    speak("Voice not detected. Default English selected")
    language_mode = "english"
    return
if "tamil" in lang:
    language_mode = "tamil"
    speak("Tamil selected")
else:
    language_mode = "english"
    speak("English selected")
def get_destination():
    global destination
    while True:
        if language_mode == "tamil":
            speak("Destination sollunga")
        else:
            speak("Say Destination")
        text = listen()
        if text is None:
            speak("Voice not detected. Please type destination")
            text = input("Enter destination: ")
        try:
            geo = Nominatim(user_agent="blind_nav_app_unique_123", timeout=10)
            place = geo.geocode(text + ", Tamil Nadu, India")
            if place is None:
                speak("Location not found. Try again")
                continue
            destination = (place.latitude, place.longitude)
            speak(f"{text} destination set")
            print("📍 Destination:", destination)
            break
        except GeocoderServiceError:
            speak("Network busy. Waiting and retrying")
            time.sleep(3)
def get_turn(p1, p2, p3):
    a = np.array(p2) - np.array(p1)

```

```

b = np.array(p3) - np.array(p2)
cross = np.cross(a, b)
if cross > 0:
    return "left"
elif cross < 0:
    return "right"
else:
    return "straight"
def calculate_bearing(lat1, lon1, lat2, lon2):
    """Calculate the bearing between two points"""
    lat1, lon1, lat2, lon2 = map(np.radians, [lat1, lon1, lat2, lon2])
    dlon = lon2 - lon1
    x = np.sin(dlon) * np.cos(lat2)
    y = np.cos(lat1) * np.sin(lat2) - np.sin(lat1) * np.cos(lat2) * np.cos(dlon)
    bearing = np.degrees(np.arctan2(x, y))
    return (bearing + 360) % 360
def bearing_to_direction(bearing):
    print("\n" + "="*70)
    print("🗺️ BLIND NAVIGATION SYSTEM - CLICK A BUTTON TO START")
    print("="*70)
    print(f"\n📱 Open in your mobile browser:")
    print(f" {protocol.upper()}: {protocol}://localhost:5000/frontend/index.html")
    if ip_address != "YOUR_IP":
        print(f" OR: {protocol}://{ip_address}:5000/frontend/index.html")
    print(f"\n🔧 Flow:")
    print(f" 1. Click 'Start Voice Assistant'")
    print(f" 2. Say 'Tamil' or 'English'")
    print(f" 3. Say your destination")
    print(f" 4. Map loads with route! 🗺️")
    print("\n" + "="*70 + "\n")
    # Wait for GPS (optional - can start without mobile)
    print("⌚ Waiting for requests...(Press Ctrl+C to stop)")
    @app.route("/navigate", methods=["POST"])
    def get_navigation():
        # Use GPS location if available, otherwise use provided or default
        if current_location:

```

```

lat, lon = current_location
print(f"\n📍 Route request: {destination_name}")
print(f"   From GPS: ({lat}, {lon})")
else:
lat = data.get("latitude", 13.0827)
lon = data.get("longitude", 80.2707)
print(f"\n📍 Route request: {destination_name}")
print(f"   From (no GPS): ({lat}, {lon})")
current_pos = (lat, lon)
if not destination_name:
return jsonify({"error": "Destination required"}), 400
# Calculate total distance
for i in range(len(route)-1):
p1 = (G.nodes[route[i]]["y"], G.nodes[route[i]]["x"])
p2 = (G.nodes[route[i+1]]["y"], G.nodes[route[i+1]]["x"])
total_distance += geodesic(p1, p2).meters
# Fallback: calculate simple distance
dist = geodesic(current_pos, dest_coords).km
time_min = (dist / 1.0) * 60 # Assuming 1 km/hour
# Generate better fallback steps
fallback_steps = generate_fallback_steps(current_pos, dest_coords,
destination_name, dist)
return jsonify({
"destination": destination_name,
"distance": dist,
"duration": time_min,
"steps": fallback_steps,
"route_coords": [list(current_pos), list(dest_coords)]
})
except Exception as e:
logger.error(f"Navigation error: {e}")
return jsonify({"error": str(e)}), 500

```

APPENDIX 2

SCREENSHOTS

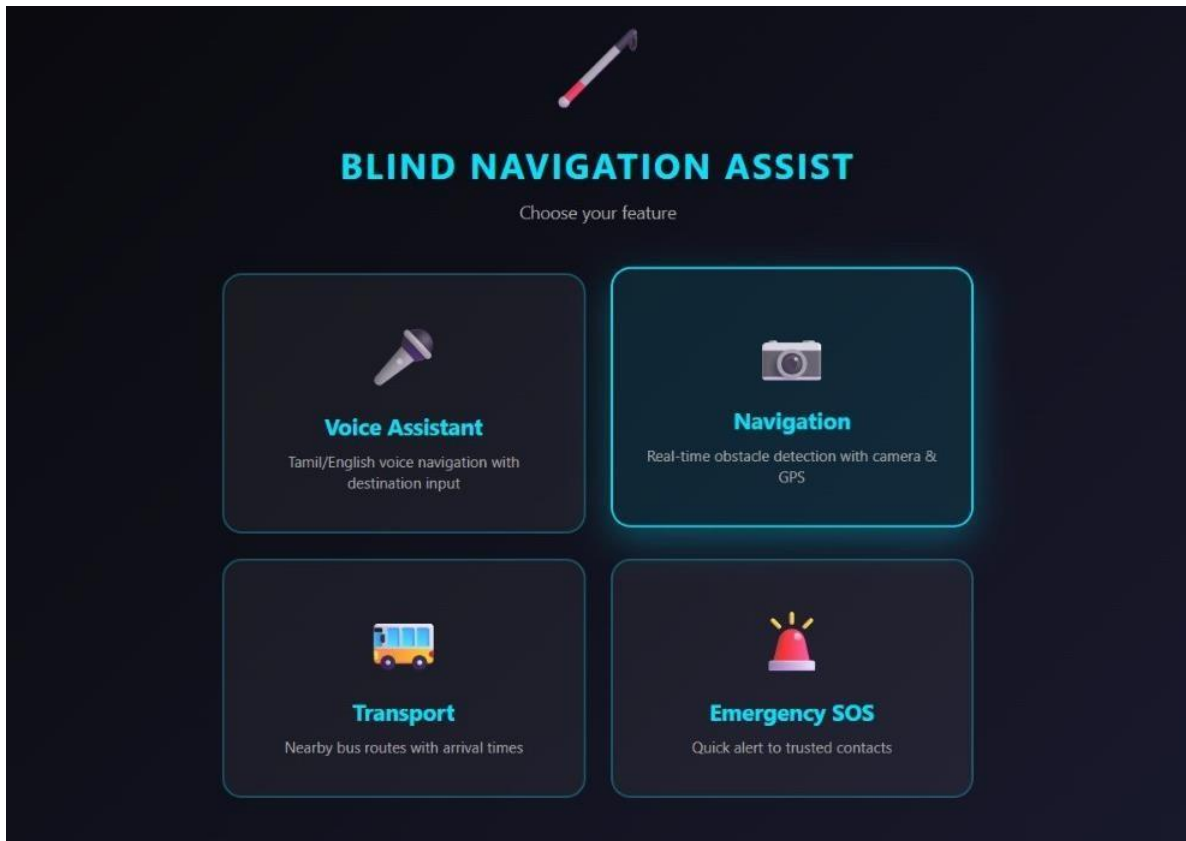


Figure:1 User Interface of Blind Navigation Assist System

The figure shows the main user interface, designed for easy, accessible interaction. The screen presents four key features arranged in a simple grid layout: Voice Assistant, which enables Tamil/English voice commands for destination input and guidance; Navigation, which uses camera and GPS for real-time obstacle detection; Transport, which provides nearby bus routes and arrival information; and Emergency SOS, which allows users to alert trusted contacts in emergencies quickly. The interface uses clear icons, highlighted sections, and a dark theme to improve visibility and usability, making it suitable for visually impaired users.

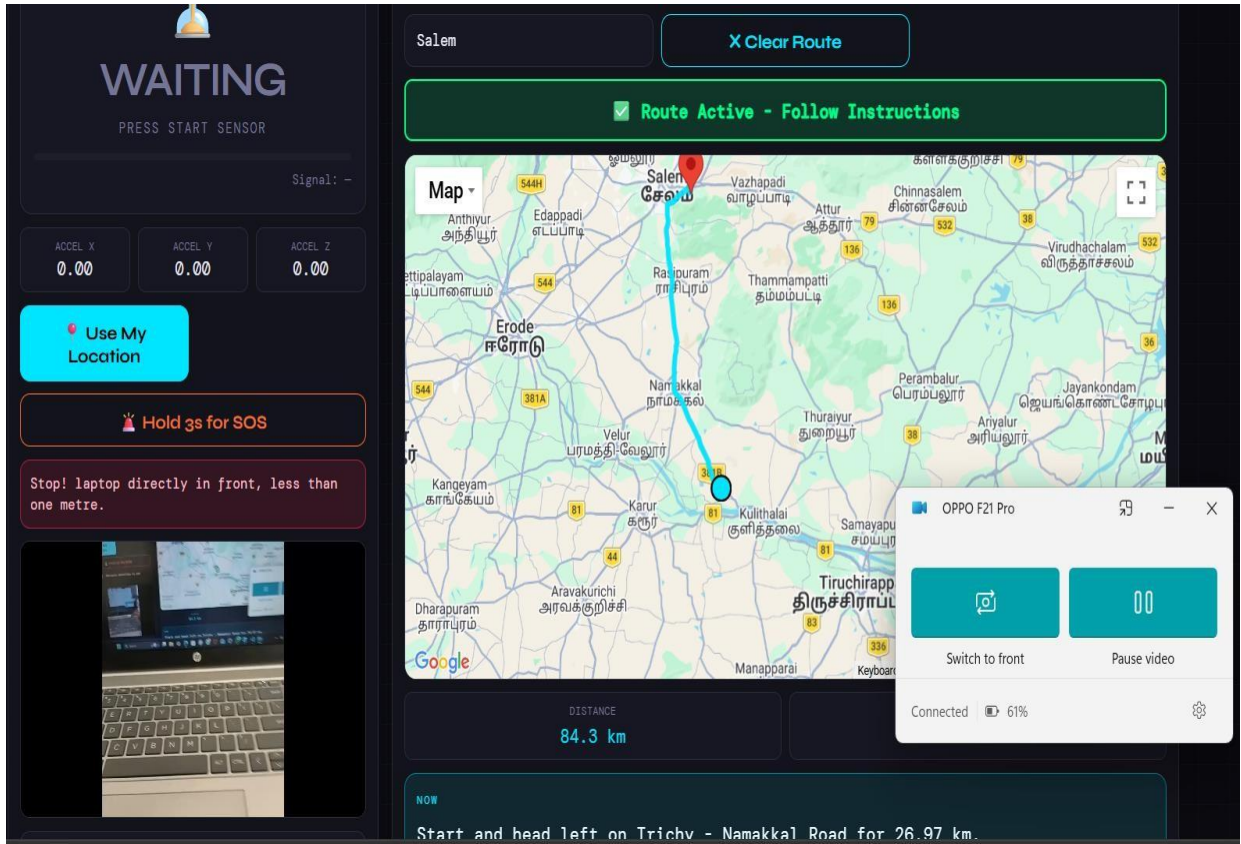


Figure: 2 System Dashboard Showing Navigation and Obstacle Alerts

The figure shows the working dashboard of the system displaying real-time navigation and obstacle detection features. On the left side, sensor data such as acceleration values (X, Y, Z) and system status are presented along with controls like “Use My Location” and sensing activation. A warning message is displayed indicating an obstacle (“traffic light directly in front, less than one metre”), demonstrating real-time object detection capability. On the right side, a map interface shows the current route with location tracking, along with distance (41.6 km) and estimated travel time (39 minutes). Below the map, step-by-step navigation instructions are provided, guiding the user through the route. The interface also includes live system logs at the bottom, indicating continuous data processing, making the system effective for safe and guided navigation.

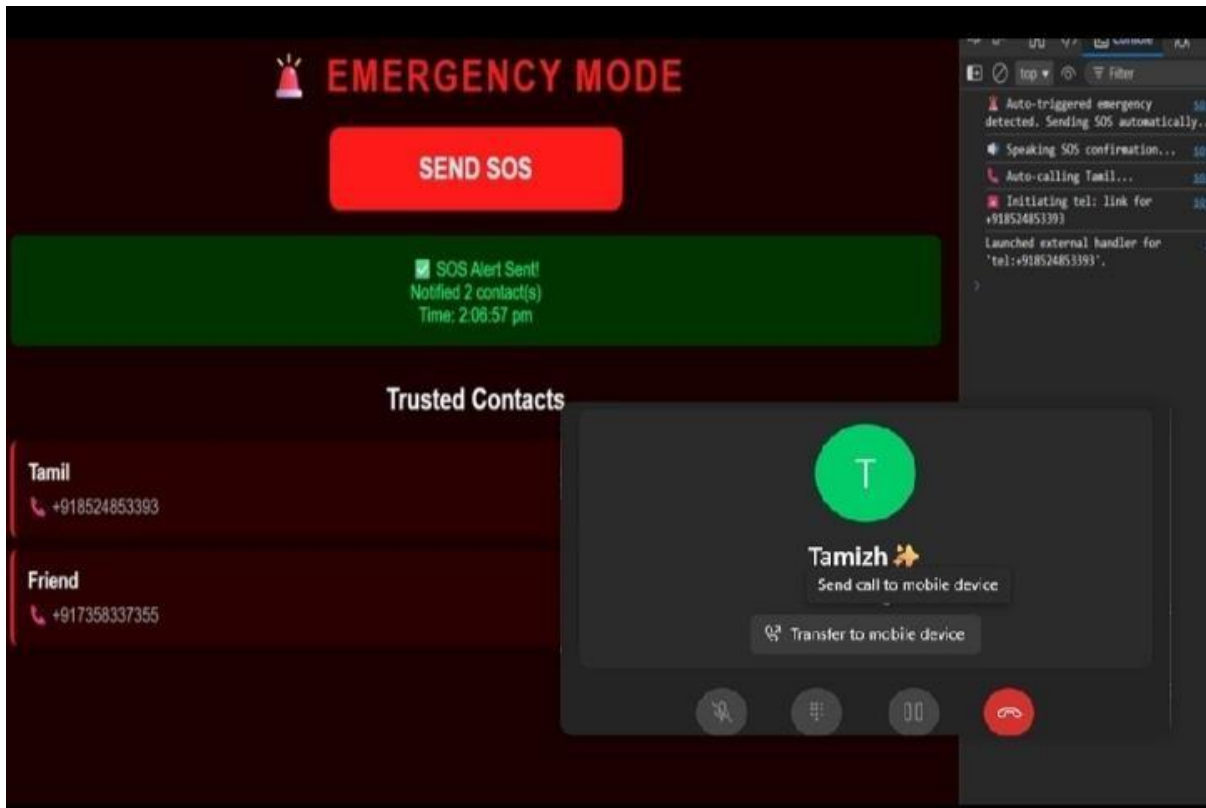


Figure:3 Emergency Alert and SOS Notification Interface

The figure shows the Emergency Mode interface of the system, which is designed to provide immediate assistance during critical situations. It includes a prominent “SEND SOS” button that allows the user to manually trigger an alert, while the system can also automatically detect emergencies and send notifications. Once activated, a confirmation message is displayed indicating that the SOS alert has been successfully sent to trusted contacts along with the timestamp. The interface also displays a list of pre-registered trusted contacts who receive the alert messages. In addition, the system demonstrates an automatic calling feature, where it initiates a call to a selected contact to ensure quick communication. Overall, this module integrates real-time alerting, messaging, and calling functionalities to enhance user safety and ensure rapid response during emergencies.

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PUBLICATIONS

SUGANYA S, NIVETHA V (621322104065), SIVA BHARATHI A (621322104100) and TAMIL K (621322104113) has published a paper entitled **“REAL-TIME AI NAVIGATION AND OBSTACLE DETECTION FOR BLIND USERS”** in International Journal of Science Engineering, and Technology in Volume 14, Issue 01, Jan-Feb-2026.

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